Cell: The Unit of Life — Class 11 NCERT One-Page Revision Notes

1. Introduction & Cell Theory

- **Definition:** The structural and functional unit of all living organisms.
- **Cell Theory:** Proposed by **Matthias Schleiden** (botanist, 1838) and **Theodor Schwann** (zoologist, 1839).
    - *Modified by* **Rudolf Virchow** (1855) — *Omnis cellula e cellula* (all cells arise from pre-existing cells).

2. Overview of Cell Types

- **Prokaryotic Cells:**
    - Lack a true membrane-bound nucleus and distinct organelles.
    - *Examples:* Bacteria, Blue-green algae, Mycoplasma, PPLO.
    - *Features:* Circular naked DNA (nucleoid), 70S ribosomes, plasma membrane invaginations called **mesosomes** (help in respiration, cell wall formation, DNA replication).
- **Eukaryotic Cells:**
    - Possess a well-organized membrane-bound nucleus and membrane-bound organelles.
    - *Examples:* Plants, animals, fungi, protists.
    - *Features:* 80S ribosomes in cytoplasm (70S in mitochondria/chloroplasts), complex cytoskeleton.

3. Cell Membrane & Cell Wall

- **Fluid Mosaic Model:** Proposed by **Singer and Nicolson** (1972).
    - Composed of a phospholipid bilayer with embedded integral and peripheral proteins.
    - *Fluid nature* allows lateral movement of proteins, crucial for cell growth, endocytosis, and cell division.
- **Cell Wall (Plants & Fungi):**
    - Rigid outer covering.
    - *Plant cell wall:* Made of cellulose, hemicellulose, pectins, and proteins.
    - *Middle lamella:* Calcium pectate layer that glues neighboring plant cells together.

4. Endomembrane System

(Coordinated functions of organelles: Endoplasmic Reticulum, Golgi Apparatus, Lysosomes, Vacuoles)

- **Endoplasmic Reticulum (ER):**
    - **RER (Rough ER):** Studded with ribosomes; active in protein synthesis and secretion.
    - **SER (Smooth ER):** Lacks ribosomes; site for lipid synthesis and steroid hormone production.
- **Golgi Apparatus:**
    - Discovered by Camillo Golgi.
    - Consists of flat, disk-shaped sacs called **cisternae** (cis and trans faces).
    - *Function:* Packaging materials (proteins/lipids) and modifying them; formation of glycoproteins and glycolipids.
- **Lysosomes:**
    - Membrane-bound vesicular structures rich in hydrolytic enzymes (hydrolases: lipases, proteases, carbohydrases) active at acidic pH. Known as "suicide bags."
- **Vacuoles:**
    - Membrane-bound space in the cytoplasm containing water, sap, and excretory products.
    - The outer membrane is called the **tonoplast**.

5. Powerhouse & Plastids

- **Mitochondria:**
    - Double-membrane bound (outer smooth, inner folded into **cristae**).
    - Site of aerobic respiration (ATP production).
    - Contains its own circular DNA, RNA, and 70S ribosomes (semi-autonomous organelle).
- **Plastids (Plant cells only):**
    - **Chloroplasts:** Contain chlorophyll and carotenoids; capture light energy for photosynthesis. Have double membranes, a fluid stroma, and stacked thylakoids (**grana**).
    - **Chromoplasts:** Fat-soluble carotenoid pigments giving yellow/orange/red colors.
    - **Leucoplasts:** Colorless plastids for nutrient storage (Amyloplasts = starch, Elaioplasts = oils, Aleuroplasts = proteins).

6. Ribosomes & Cytoskeleton

- **Ribosomes:** Non-membrane-bound granular structures composed of RNA and proteins.
    - **70S** (in prokaryotes, mitochondria, chloroplasts) = 50S + 30S subunits.
    - **80S** (in eukaryotes) = 60S + 40S subunits.
- **Cytoskeleton:** Network of filamentous proteinaceous structures (microtubules, microfilaments) providing mechanical support, motility, and maintenance of cell shape.
- **Cilia and Flagella:** Hair-like outgrowths of cell membrane. Core is called the **axoneme** (9 + 2 array of microtubules).
- **Centrosome & Centrioles:** Centrosome contains two cylindrical structures called centrioles (arranged at right angles, cartwheel structure with **9 + 0** microtubule

arrangement) that form the basal body of cilia/flagella and spindle apparatus during cell division.

7. The Nucleus

- Discovered by **Robert Brown** (1831).
- **Nuclear Envelope:** Double-membrane with nuclear pores for RNA and protein transport between nucleus and cytoplasm.
- **Nucleoplasm / Chromatin:** Contains loose network of nucleoprotein fibers called chromatin, DNA, and histones.
- **Nucleolus:** Non-membrane-bound spherical body inside the nucleus; active site for **rRNA synthesis**.

Quick Revision Check:

- Can you differentiate between 70S and 80S ribosomes?
- What is the difference in microtubule arrangement between an axoneme (9+2) and a centriole (9+0)?
- What are the functions of the cis and trans faces of the Golgi body?